UNITY XR ENGINEER

BRIDGING ART AND ENGINEERING

LOÏCK RIVEMALE

portfolio

LinkedIn

GitHub

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EXPERIENCE

8K AR Video Platform - *Unity XR Engineer* ***(NDA client)***

December 2025 - Present

- Developed a custom video encoding/decoding pipeline to compress 8K source content into a 4K-compatible mobile delivery format.
- Built a Unity Android AR application for AR glasses, with a companion updater app, custom media processing tools, and backend content delivery system.
- Owned the technical architecture, development, maintenance, deployment, and client-facing delivery across the project.

RETìníZE, Belfast, UK - *Unity XR Engineer*

November 2020 - March 2025

Core member of the engineering team on Animotive, a VR multiplayer virtual production and animation platform, from pre-release development through commercial launch and post-release support.

- Developed a GPU-based, collider-less raycast system performing thousands of submesh-precision ray tests in a single draw call, replacing per-ray CPU physics queries.
- Wrote all the in-house custom shaders and render features while optimizing the rendering pipeline for complex VR scenes.
- Built and maintained a complete UI toolkit for VR.
- Took part in evaluating and integrating full-body tracking systems, including Vive trackers and Xsens.

Airbus Defense & Space, Toulouse, France - *Unity VR developer*

July 2017 - June 2020

Core member of Airbus Group's internal VR visualizer (**DragonFly**) development team from prototyping to post-release.

- Built a procedural 3D shape generation library and two TCP interfaces in C# and Java to connect DragonFly with external software.
- Developed a double-precision solar system simulation and satellite telemetry visualizer to support DragonFly's adoption by Toulouse Mission Control center.
- Managed client relationships, roadmaps, and demos for projects integrating DragonFly into other teams.

VisioPM, Toulouse, France - *Unity C# developer intern*

April 2016 - August 2016

- Built a proof of concept for a Unity-based autostereoscopic rendering engine.

XNoise – *Free open-source side project*

2025

- Ported LibNoise into a node-based Unity noise generation tool with CPU/GPU execution and planar, spherical, and cylindrical map export.

STUDIES

Epitech, France Master's degree in software engineering
2013 – 2018

Beijing Jiaotong university, Beijing, China - Computer science
2016 – 2017

PRIMARY SKILLS

- **C#**
- **Shaderlab (HLSL)**
- **Shader graph**

- **Unity (URP, HDRP, custom RP)**
- **Git**
- **Photon Bolt**

- **Virtual reality**
- **Augmented reality**
- **Mathematics**
- **GPU acceleration**
- **Render features**
- **Editor tooling**

WORKING KNOWLEDGE

- **C**
- **C++**
- **Java**
- **Javascript**

- **Blender**
- **Photoshop/Photopea**
- **Gaea**
- **Linux**

- **TCP/UDP networking**
- **Protocol buffer**
- **WebGL**
- **SQL**
- **WPF**

LANGUAGES

English : Bilingual

French : Native